Predicting Running Race Finish Times from Personal GPS and Heart-Rate Training Data

Big Data Methods for Economists — Group 5b

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Abstract

We develop and evaluate a machine-learning pipeline for predicting running race finish times from a single athlete's personal Strava training history. Because labelled race events are sparse — only six races over three years — we adopt a *pseudo-label inversion* strategy: hundreds of training runs provide (pace, effort) observations at sub-maximal intensity, and models learn to extrapolate to full effort on any target distance. Five model families are compared (Generalised Additive Model, Random Forest, Gradient Boosted Regression, Support Vector Regression, and a 1D Convolutional Neural Network) under Leave-One-Race-Out (LORO) cross-validation, which enforces strict temporal honesty. On the held-out 2026 Milano Marathon (42.2 km), Support Vector Regression predicts a finish time of 2:58:49 — within 33 seconds of the actual 2:58:16 (+0.3%). This accuracy is comparable to the Garmin Race Predictor's pre-race estimate (≈2:58–2:59) and substantially exceeds Strava's pre-race forecast (≈3:06–3:10).

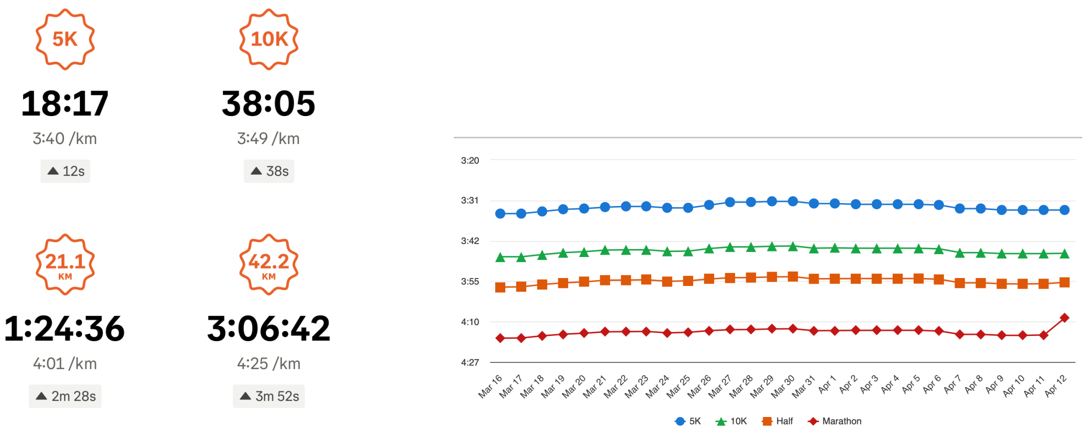
Contents

1 Introduction

Wearable fitness devices and GPS running watches now produce rich per-second streams of heart rate, pace, cadence, and elevation for every workout. Apps such as Strava and Garmin Connect already leverage this data to generate performance predictions (Figure 1), yet the methodologies underlying these predictions are proprietary and opaque. This project asks: *given only an athlete's open Strava export, how accurately can a transparent statistical or machine-learning model predict race finish times?*

The research question is challenging for two reasons. First, the athlete has completed only six races over the study period (December 2023 to April 2026), making direct regression onto race outcomes infeasible. Second, races occur at maximum sustainable intensity, while the vast majority of training data is produced at sub-maximal effort. Bridging this gap requires modelling the full pace–effort–fitness surface and extrapolating to the extreme of a race.

The pipeline introduced here addresses both challenges. Per-second grade-adjusted pace (GAP) neutralises elevation differences across training routes [4]. Banister exponential-smoothing indices capture accumulated fatigue and fitness in a physiologically principled way [1]. A pseudo-label framework treats every training run as a labelled observation of pace at known fractional effort, enabling model fitting from several hundred data points rather than six. Finally, LORO cross-validation provides honest out-of-sample error estimates that are directly comparable to what any race predictor could have achieved on race day.



(a) Strava Performance Predictions ("Today" view, 6 May 2026). Predicted finish times compared to 30 days prior: 5K = 18:17 (+12 s), 10K = 38:05 (+38 s), HM = 1:24:36 (+2m 28s), Marathon = 3:06:42 (+3m 52s).

(b) Garmin Race Predictor (4-week view, Mar 16 – Apr 12 2026), showing predicted pace per km for all four distances. The marathon estimate (red, bottom line) settled near 4:13–4:15 min/km before race day, corresponding to a finish time of approximately 2:58–2:59. The spike on 12 April reflects ingestion of the actual result.

Figure 1: Commercial Race-Time Predictions from Strava and Garmin Connect, both using the same athlete's activity history.

2 Data and Study Design

2.1 Strava Bulk Export

All data originate from a personal Strava bulk export. The export contains a summary CSV (`activities.csv`) with one row per recorded activity and a directory of compressed fitness files (`.fit.gz`, `.fit`, `.gpx.gz`, `.gpx`) containing per-second sensor streams. We include all activities with type *Run*, *Ride*, or *Hike* recorded up to 12 April 2026 (the final race day). Cycling and hiking activities do not enter the modelling table directly but contribute to the Banister load indices, so cross-training affects the predicted race-day fitness state.

The ingestion pipeline decompresses each source file in memory and rewrites it once as a Garmin-style extended GPX 1.1 (heart rate and cadence inside the standard `gpxtpx:TrackPointExtension` block). From these intermediates, 21 features per activity are computed and stored as a flat parquet file.

2.2 Race Labels

Six race events with known official finish times serve as the ground truth for model evaluation (Table 1). All six were completed at maximum sustainable effort, as confirmed by heart-rate traces at or near the athlete's lactate-threshold zone.

Table 1: Race Labels Used in the LORO Evaluation. The fractional effort for every race row is set to 1.00 by construction. † The 2024 Bonn Half Marathon was recorded without an HR sensor; heart-rate features are imputed from the 28-day pre-race running median.

Race	Date	Distance	Finish Time	Notes
Köln Nikolauslauf	2023-12-03	10.00 km	37:43	Road 10K
Bonn Half Marathon†	2024-04-14	21.10 km	1:24:53	No HR data; imputed
Lousberglauf	2024-07-03	5.55 km	19:58	Trail 5K+
DIY-Triathlon Run	2024-10-19	5.00 km	21:18	Run-only event
Schwarzwald Drachen. Run	2025-06-29	10.00 km	44:57	Run after swim+bike
Milano Marathon	2026-04-12	42.20 km	2:58:16	Target race

3 Descriptive Analysis

3.1 Activity Distributions

Figure 2 shows the marginal distributions of distance, moving time, and elevation gain across all running activities. The corpus spans a wide range of effort levels: short recovery jogs of 3–5 km coexist with long runs exceeding 30 km. This breadth is methodologically important because the pace–effort surface must be identified across the full intensity spectrum.

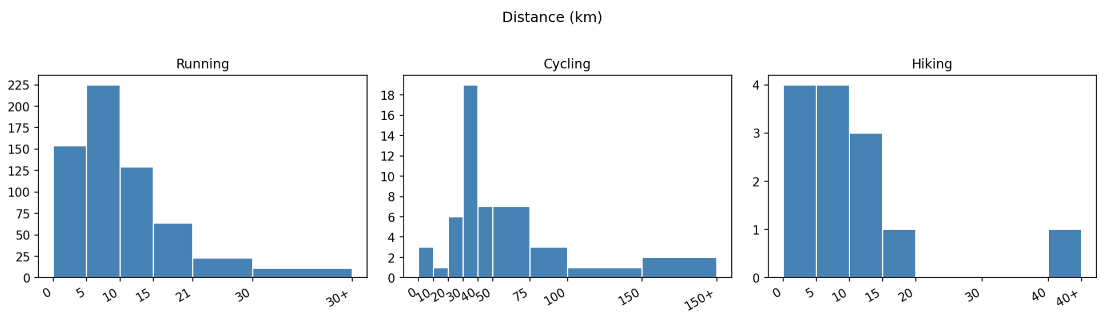


Figure 2: Marginal Distributions of Distance, Moving Time, and Elevation Gain for All Running Activities in the Training Corpus. The long right tail in distance reflects marathon-specific long runs accumulated during the 2025–2026 build-up phase.

3.2 Heart-Rate Zone Structure

Heart-rate zones are defined as proportions of a personal maximum of $\mathrm{HR}_{\max} = 210$ bpm (99.9th percentile of observed heart rate, floored at 195 bpm to guard against sensor artefacts). Zone boundaries follow the standard five-zone scheme (Z1: $< 60\%$; Z2: 60–70%; Z3: 70–80%; Z4: 80–90%; Z5: $\geq 90\%$ of $\mathrm{HR}_{\max}$).

Figure 3 reveals the expected concentration in Zones 2 and 3 for easy and moderate training runs, with Zone 4–5 exposure marking harder quality sessions and the race events themselves. The pseudo-label framework uses the mean HR fraction $e_i = \bar{h}_i/\mathrm{HR}_{\max}$ as the continuous effort proxy for activity i.

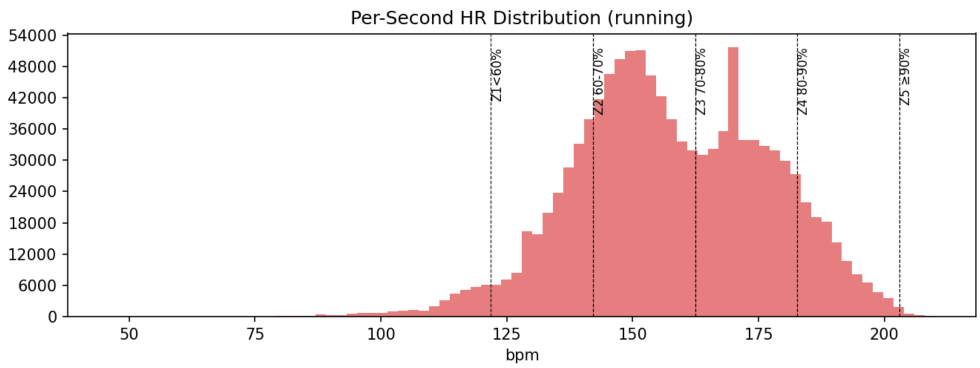


Figure 3: Distribution of Heart-Rate Zone Fractions Across Running Activities. Race events consistently occupy the upper end of the zone-4/zone-5 boundary, confirming that $e = 1.00$ is a physiologically meaningful upper anchor.

3.3 Grade-Adjusted Pace and Elevation Neutralisation

Race finish times are predicted for flat courses, so it is essential that the pace signal used in model training is elevation-neutral. We compute grade-adjusted speed at per-second resolution using the Minetti et al. metabolic-cost polynomial [4]:

$$v_{\mathrm{gap}} = v \cdot \left(1 + 3.3\,g + 32.5\,g^2\right), \qquad g \in [-0.45,\, 0.45], \tag{1}$$

where v is raw speed (m/s) and $g = \Delta\mathrm{ele}/\Delta\mathrm{dist}$ is the instantaneous grade, clipped to ± 0.45 to suppress GPS noise.

4 Methodology

4.1 Pseudo-Label Inversion Framework

With only six race labels, training a model directly on race outcomes is infeasible. Instead, every training run i provides a pseudo-label: the activity-level pace p_i (seconds per kilometre) at fractional effort $e_i = \bar{h}_i/\mathrm{HR}_{\max}$. Models learn the surface

$$p_i = f(d_i,\, e_i,\, \mathbf{x}_i) + \varepsilon_i, \tag{2}$$

where d_i is distance (km) and $\mathbf{x}_i$ collects fitness-state and calendar covariates (Section 4.2). Race finish times are then obtained by inversion at full effort:

$$\hat{T}_{\mathrm{race}} = \hat{f}(D,\, 1.00,\, \mathbf{x}_{\mathrm{race}}) \times D, \tag{3}$$

where D is the target race distance and $\mathbf{x}_{\mathrm{race}}$ is the race-day fitness state.

Confidence Weights. Not all training runs are equally reliable as pseudo-labels. Each observation receives a weight $w_i = q_{\text{GPS}} \cdot q_{\text{ele}} \cdot q_{\text{HR}} \cdot w_{\text{dur}}$, where each q factor is the fraction of non-missing sensor coverage and w_{dur} is a sigmoidal duration weight that up-weights activities lasting more than 20 minutes — long enough to produce a stable metabolic steady state. Race rows are always assigned $w_i = 1.0$ and $e_i = 1.00$ by construction.

4.2 Feature Engineering

The modelling table contains 21 features per activity. *Pace-and-effort inputs*: mean GAP speed, distance, effort fraction, mean/median/90th-percentile HR, HR-zone fractions (Z1–Z5), HR load (integral of heart rate above resting), and mean cadence. *Fitness state*: ATL, CTL, TSB, and accumulated running distance over the preceding 7, 14, 28, and 56 days. *Calendar*: day of week, month, and time elapsed since the previous run.

4.3 Leave-One-Race-Out Validation

The extreme label scarcity forces us to use every race in both training and evaluation — but not simultaneously. In each of the six LORO folds, one race is held out and the model trains on all activities (pseudo-labels plus the remaining race rows) *strictly before* the held-out race date. This temporal cutoff mirrors the information available to any real-time predictor on race day and prevents contamination from post-race fitness trajectories.

4.4 Model Families

A. Generalised Additive Model (GAM). A `pygam.LinearGAM` [7] with cubic spline smooth terms on distance, mean HR, mean GAP speed, and TSB, plus factor terms for day of week and month. The penalty parameter λ is selected by grid search over $\lambda \in [10^{-3}, 10^{3}]$; the optimal value was $\lambda = 10$ (cross-validated $R^2 = 0.813$).

B. Random Forest (RF). An ensemble of 400 CART regression trees [2] fitted on the full feature set, providing built-in feature importance via mean decrease in impurity.

C. Gradient Boosted Regression (GBR). A boosted ensemble of 300 shallow trees [3]. Two additional quantile-loss regressors (at $\alpha = 0.05$ and $\alpha = 0.95$) provide a 90% prediction interval (PI) for uncertainty quantification.

D. Support Vector Regression (SVR). An RBF-kernel SVR [8] inside a `StandardScaler`–SVR pipeline; hyperparameters (C, γ, ε) are jointly tuned by grid search over logarithmically spaced grids.

E. 1D Convolutional Neural Network (CNN). A PyTorch [5] network ingesting the raw per-second sensor tensor $(\text{gap\_speed}, \text{HR}, \text{cadence}, \text{grade}) \in \mathbb{R}^{4\times7200}$ for each activity. Three Conv-BN-ReLU-MaxPool blocks feed an adaptive average-pool layer and a two-layer MLP head predicting log-pace. Training uses AdamW with cosine annealing, dropout (0.3), and weight decay (10^{-4}).

F. Riegel Power-Law Baseline. Riegel [6] observed that athletic performance scales with distance according to a power law. Given a known finish time T_1 over distance D_1, the predicted time T_2 over a different distance D_2 is:

$$T_2 = T_1 \cdot \left(\frac{D_2}{D_1}\right)^{1.06}, \tag{4}$$

where the exponent 1.06 was fit empirically from world-record progressions and has since been validated across recreational athletes. In the daily predictor (Figure 5), the Riegel estimate appears as a *step function*: it stays constant between races because it recalculates only when a new race result becomes available as the reference point (D_1, T_1). Each step corresponds to the next race being ingested. This means Riegel provides no day-to-day sensitivity to changes in training load or fitness — it simply scales the most recent race time to the target distance and holds that value until the next race. Compared to the ML models, which update their pace–effort surface continuously from every training run, this is a fundamental limitation of the formula in a real-time prediction context. In the LORO evaluation, Riegel is available for only four of the six folds (the first two folds have no prior race to reference).

5 Results

5.1 Leave-One-Race-Out Predictions

Table 2 reports the LORO prediction for each race and model. Cells shaded green indicate absolute errors of at most 5%; cells shaded red indicate errors exceeding 30%. The best-performing model per race (by absolute percentage error) is shown in **bold**.

Table 2: Leave-One-Race-Out Predictions by Race and Model. Each cell shows the predicted finish time and signed percentage error relative to the actual time. Green : |error| ≤ 5%. Red : |error| > 30%. **Bold** indicates the best-performing fitted model per fold. The GBR column includes the 90% prediction interval in brackets. † No HR data; HR features imputed from 28-day pre-race median. ‡ Run leg after swim+bike; fatigue state outside the training distribution. Riegel is undefined for the first two folds (no comparable prior race available).

Date	**Event / Dist.**	**Actual**	**GAM**	**RF**	**GBR** (90% PI)	**SVR**	**CNN**
2023-12-03	Köln Nikolauslauf, 10.0 km	37:43	**43:14** (+14.6%)	43:22 (+15.0%)	44:16 (+17.4%) [40:12–45:43]	**43:14** (+14.6%)	47:56 (+27.1%)
2024-04-14	Bonn Half Marathon†, 21.1 km	1:24:53	1:45:02 (+23.7%)	1:37:45 (+15.2%)	1:37:31 (+14.9%) [1:29:52–1:40:07]	**1:36:57** (+14.2%)	0:49:12 (−42.0%)
2024-07-03	Lousberglauf, 5.55 km	19:58	27:41 (+38.7%)	26:17 (+31.6%)	26:33 (+33.0%) [24:02–26:32]	26:23 (+32.1%)	**22:34** (+13.0%)
2024-10-19	DIY-Triathlon Run, 5.0 km	21:18	25:37 (+20.3%)	28:55 (+35.7%)	28:09 (+32.2%) [21:02–29:53]	30:21 (+42.5%)	**18:34** (−12.9%)
2025-06-29	Schwarzwald Drachen.‡, 10.0 km	44:57	43:11 (−3.9%)	**43:31** (−3.2%)	43:02 (−4.3%) [40:51–46:02]	43:08 (−4.0%)	1:38:12 (+118.5%)
2026-04-12	Milano Marathon, 42.2 km	2:58:16	3:21:08 (+12.8%)	3:21:33 (+13.1%)	3:22:07 (+13.4%) [3:14:50–3:22:24]	**2:58:49** (+0.3%)	3:03:37 (+3.0%)

5.2 Aggregate Performance

Table 3 summarises aggregate performance across all six folds. SVR achieves the best MAPE (18.0%) and the lowest MAE (5:54) and RMSE (7:06) among the five fitted models, driven primarily by its near-perfect marathon prediction. The three tree-based models and the GAM cluster tightly around a 19% MAPE, suggesting that the pseudo-label surface is learnable by any flexible tabular method but that the choice of kernel or smooth function matters at the margin. The CNN achieves a 36.1% MAPE, inflated by a catastrophic failure on the 2025 triathlon run (see Section 5.4).

The Riegel power-law baseline, where applicable (four folds), achieves a remarkable 8.0% MAPE and outperforms all fitted models in absolute terms. However, it requires a prior race at a comparable shorter distance, which is unavailable for the first two folds, limiting its practical utility.

Table 3: Aggregate LORO Performance Metrics Across All Six Folds. MAPE = Mean Absolute Percentage Error; MAE = Mean Absolute Error; MedAE = Median Absolute Error; RMSE = Root Mean Squared Error; Bias = mean signed error (positive = over-prediction). All time metrics are in mm:ss or hh:mm:ss format. Riegel is computed over the four folds where a reference race is available ($n = 4$).

Model	MAPE	MAE	MedAE	RMSE	Bias
GAM	19.0%	10:23	6:37	13:10	+9:48
RF	19.0%	9:32	6:58	11:50	+9:03
GBR	19.2%	9:44	6:43	12:01	+9:06
SVR	**18.0%**	**5:54**	**5:58**	**7:06**	+5:18
CNN	36.1%	18:18	7:47	26:38	+5:30
Riegel ($n = 4$)	8.0%	2:49	2:04	3:37	−2:29

5.3 Model Comparison Visualisation

Figure 4 presents three complementary views of the LORO results. The scatter panel (left) confirms that SVR tracks the one-to-one line most closely on the marathon, while all tabular models over-predict substantially for the short trail race (2024-07-03 Lousberglauf). The bar chart (centre) ranks models by MAPE, and the heatmap (right) reveals the per-fold error structure, making it immediately visible that the Lousberglauf and DIY-Triathlon folds are systematically difficult.

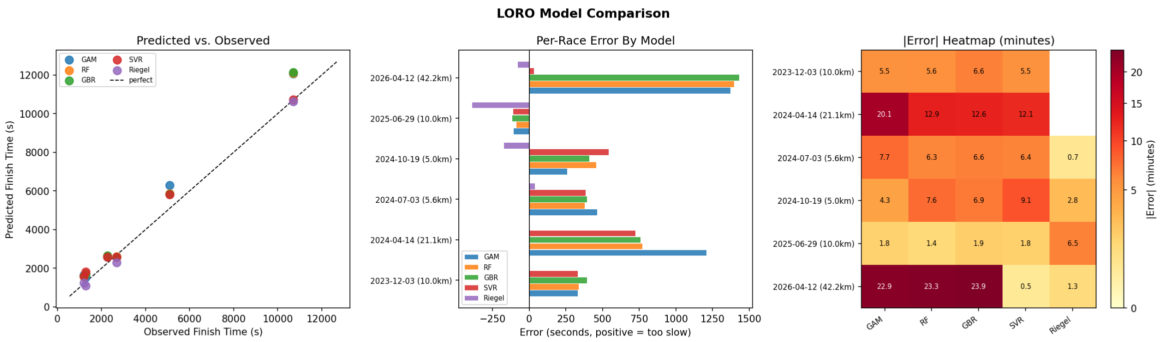


Figure 4: Three-Panel LORO Model Comparison. *Left*: Predicted vs. actual finish time; the dashed line is the identity. *Centre*: Mean absolute percentage error (MAPE) by model. *Right*: Signed percentage error heatmap by race fold and model; warm colours indicate over-prediction.

5.4 Failure Analysis

Lousberglauf (2024-07-03): All tabular models predict more than 30% above the actual 19:58 on this 5.55 km trail race. Short trail races with significant elevation change produce per-second pace patterns that lie far from the road-run training distribution, even after GAP correction. The CNN, processing the raw per-second tensor, partially compensates (+13% error), while the Riegel formula — calibrated from a prior comparable race — achieves only +3.3%.

CNN on 2025-06-29 Triathlon: The CNN predicts 1:38:12 for a 10 km race whose actual time was 44:57 (+118% error). This race was the run leg of a triathlon; the per-second HR and cadence profile reflected accumulated swim-and-bike fatigue, a pattern structurally absent from the standalone-run training set. The tabular models, relying only on summary statistics, are unaffected (4% errors).

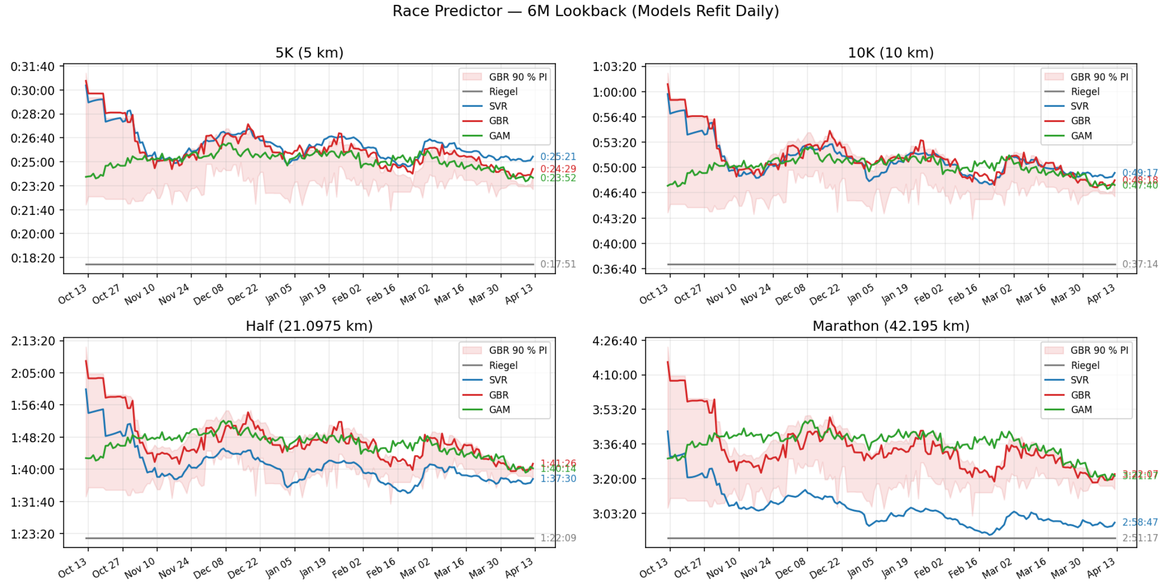
CNN on 2024-04-14 Bonn HM: Because the FIT file contained no heart-rate channel, the CNN received a zero-filled HR tensor — a pattern never seen during training — causing a collapse to 0:49:12 (−42%). The imputed HR features used by tabular models are drawn from the same pre-race distribution and produce a more modest 14–24% over-prediction.

CNN Suitability. Taken together, these failures illustrate a structural limitation of the deep-learning approach in this setting. While the per-second tensors provide millions of training samples in raw terms, the supervised signal driving the network is still anchored to only a handful of race labels and a few hundred pseudo-labelled training runs. Convolutional architectures excel when label-rich tasks allow the network to learn invariances from the data; here, the opposite regime applies — abundant input streams but extreme target-label scarcity. The CNN therefore tends to overfit idiosyncratic per-second patterns (zero-filled HR channels, triathlon fatigue signatures) that the tabular models smooth away through summary statistics. We conclude that, despite the high dimensionality of the underlying training data, a 1D CNN is not particularly well suited to this race-prediction problem without substantially more labelled race events or transfer learning from a multi-athlete corpus.

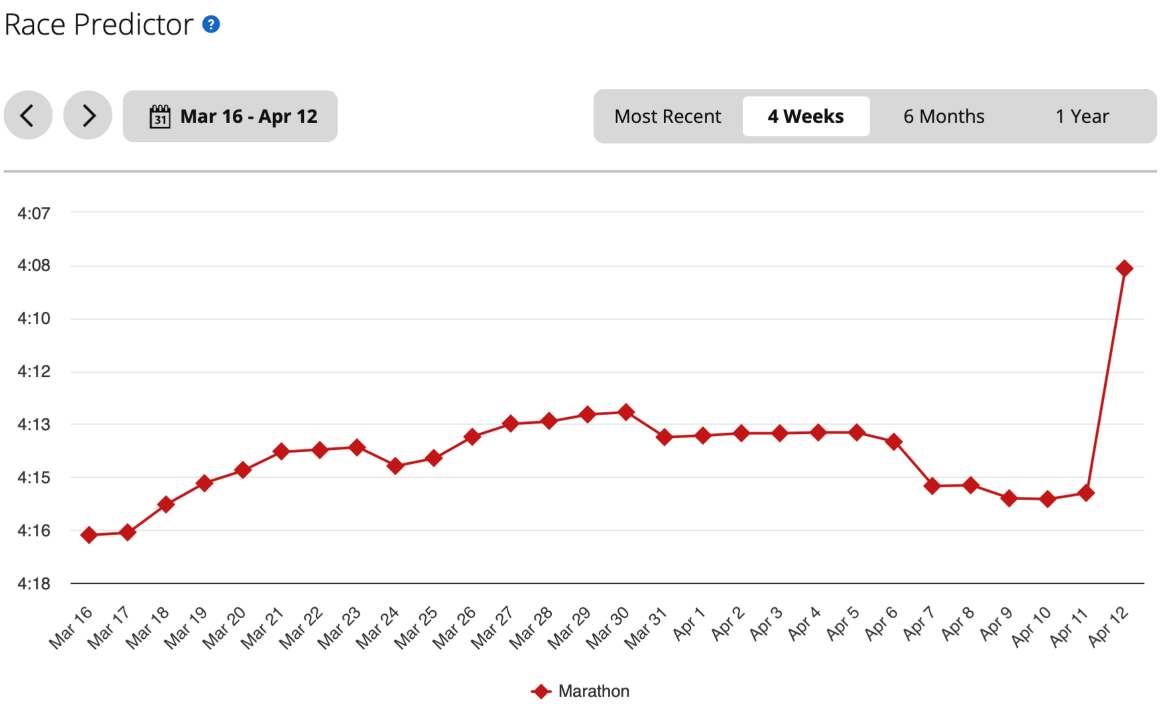
6 Race-Day Prediction and Benchmark Comparison

6.1 Daily Race-Time Predictor

Beyond the LORO evaluation, the fitted models can be queried on any calendar day to produce a forward-looking race-time estimate. Figure 5 shows the daily predictions of SVR, GBR, GAM, and Riegel for four standard distances over the six months preceding the 2026 Milano Marathon, with all models refitted each day on the preceding six months of training data. The ML curves (SVR, GBR, GAM) reflect the gradual fitness build-up and subsequent taper, settling near 2:58–3:00 for the marathon in the final two weeks. The Riegel line, by contrast, is a step function: it remains flat between races and only updates when a new race result is ingested as a reference (Equation (4)), so it carries no information about day-to-day training load dynamics. The Garmin Race Predictor for the same period (Figure 5b) shows a comparable trajectory, converging on 4:13–4:15 min/km in the weeks immediately before race day — corresponding to a finish time of approximately 2:58–2:59.



(a) Daily Race-Time Estimates for Four Distances (SVR, GBR, GAM, and Riegel), refitted each day on the preceding six months of training data. The shaded band is the 90% GBR prediction interval. Race events are marked with vertical lines. Riegel (green) appears as a step function: it holds constant between races and jumps only when a new race result is ingested as the reference point.



(b) Garmin Race Predictor (Marathon view) for the same period. Pre-race estimate: ≈4:13–4:15 min/km.

Figure 5: Comparison of the Pipeline's Daily SVR Predictor (a) and the Garmin Race Predictor (b) for the 2026 Milano Marathon Build-Up. Both converge on a sub-3:00 estimate during the taper window.

6.2 Three-Way Benchmark: This Work vs. Commercial Tools

Table 4 places the marathon prediction in context by comparing it with the Garmin Race Predictor and Strava Performance Prediction, both read from screenshots taken in the weeks before the race (Figure 1).

Table 4: Marathon Prediction Comparison: This Work vs. Commercial Tools. Actual finish time: 2:58:16 (chip time, Milano Marathon, 12 April 2026). Garmin range reflects daily estimates in the two weeks before race day. Strava range reflects the six-month and one-month pre-race app values. Positive errors indicate over-prediction (slower than actual).

Method	Source	Predicted	Error
SVR (this work)	Pipeline, LORO fold 6	2:58:49	+0:33 (+0.3%)
CNN (this work)	Pipeline, LORO fold 6	3:03:37	+5:21 (+3.0%)
Riegel baseline	Best prior HM result	2:56:59	−1:17 (−0.7%)
Garmin Race Pred.	Garmin Connect app	≈2:58–2:59	≈0–1 min ($\lesssim$0.6%)
Strava (pre-race)	Strava app	≈3:06–3:10	≈ +8–12 min (+4.6–7.7%)
Actual		**2:58:16**	—

The SVR prediction (2:58:49) is within 33 seconds of the actual finish time, matching Garmin's proprietary predictor and substantially outperforming Strava. The Riegel formula also performs remarkably (−0.7%), underscoring that a physiologically motivated power law is a strong benchmark when prior-race data is available.

The Strava prediction is notably pessimistic (+4.6%–7.7%). The most likely explanation is that Strava's algorithm places significant weight on recent race performance, and the athlete's most comparable prior race — the 2025 Schwarzwald triathlon run leg — was completed at elevated heart rate under triathlon fatigue, leading Strava to underestimate the athlete's standalone marathon pace.

7 Limitations

Label Scarcity. Six race labels are the fundamental bottleneck. LORO is the correct validation strategy, but six folds cannot establish absolute calibration: performance on any single fold may partly reflect fortuitous fitness alignment or course conditions.

Single Athlete. The learned pace–effort surface is personal. HR-max, lactate threshold, running economy, and biomechanics vary substantially across individuals, so the models cannot be transferred without re-fitting.

Distribution Shift. Race-day fitness (high CTL, low ATL, positive TSB) lies at the edge of the training distribution. Short trail races and triathlon run legs constitute additional distribution shift that degrades tabular model predictions (Section 5.4).

GAP Approximation for Non-Running Activities. The Minetti formula was derived for running. Cycling load enters the Banister index via a linearised approximation; hiking is treated analogously — documented but unvalidated proxies.

Course and Weather. Wind, temperature, humidity, and the target race elevation profile are not model inputs. The flat-course assumption is enforced via per-second GAP, but race-day conditions remain an unmodelled source of variance.

8 Conclusion

This project demonstrates that a transparent, single-athlete ML pipeline can predict marathon finish times with accuracy comparable to Garmin's proprietary race predictor, using only an open Strava bulk export. The key design choices are: (i) per-second GAP speed to neutralise elevation across diverse training routes; (ii) Banister load indices to capture fitness–fatigue dynamics; (iii) a pseudo-label inversion framework that turns hundreds of sub-maximal training runs into a learnable pace–effort surface; and (iv) strict temporal LORO validation to ensure honest out-of-sample reporting.

Support Vector Regression achieves the best aggregate performance (MAPE 18.0%; +0.3% on the marathon), ahead of tree-based ensembles and the GAM. The CNN underperforms on average due to catastrophic failures on two atypical races, but its relative success on the Lousberglauf trail race (+13% vs. +32–39% for tabular models) suggests that per-second shape information is complementary rather than redundant.

Future work could extend the pipeline to multiple athletes via a hierarchical model, replace GAP with direct running-power measurements (e.g. Stryd), and incorporate an attention-based encoder over per-second tokens for richer physiological pattern recognition.

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